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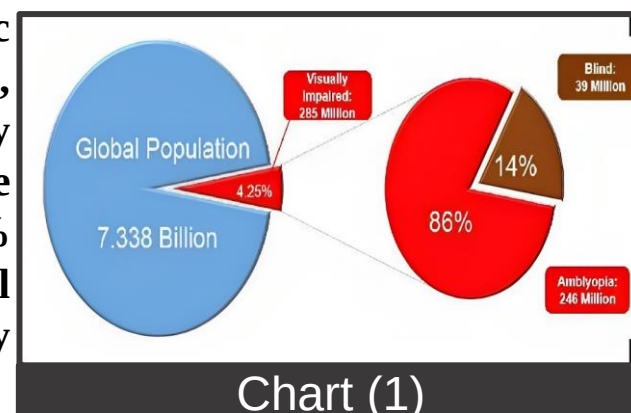


Abstract

Recently, a lot of catastrophes have arisen because of inappropriate and unplanned activities: lack of industrial bases, poor health care for disability, and many innumerable ones. However, this research focuses on different types of issues that are not related to the environment but to humans themselves. Being a hateful fate for about 285 million people around the globe, visual impairment is one of the most controversial dilemmas that has most of its cases not cured yet. Although medicine is developing all the time, it is always so complicated when it comes to visual issues. Thus, a lot of gigantic organizations are working hard supported by people all around the globe to help visually impaired people to live with the least difficulties possible. Lots of ideas have been proposed aiming mainly to make life easier for people who have serious sight issues. Sharing our part in this worldwide movement, electronic devices were manufactured by which visually impaired people can sense their surrounding objects and even read texts using Raspberry Pi-4 and ultrasonic sensors that are attached to goggles in addition to using a trained model. Also, two design requirements have been determined to assure the validity of the project, Accuracy of more than 85% and time responding between 3 and 5. The project sought to be accurate while maintaining a fast action or response. After conducting the test plan, one of the major findings was that the prototype successfully passed all the tests and achieved the specified design and solution requirements from cost, eco-friendly, Accuracy, practicality, and time response.

Introduction

It is estimated that Egypt is facing lots of Catastrophic problems (grand challenges), including pollution, health, deterioration, and lack of recycling. However, our study discusses a very specific case related to the industrial base and public health in Egypt. According to WHO, about 4.25% (285 million) of the world's population experience visual issues. Furthermore, 14% of them (39 million) are completely blind, as shown in chart (1).



That leaves us with a serious problem to deal with. Blind people are considered a non-productive class. Thus, ignoring them can significantly affect our economy. Also, there is limited access to advanced technology in the Egyptian industry. Also, there is poor production of technological products. That is why a lot of organizations are working hard to help visually impaired people live with the least difficulties possible. These challenges combined in a general challenge that includes developing an ICT (information communication technology) system to help patients exposed to chronic diseases and disabilities, at the same time improving both public health and industrial base. Many solutions have been proposed, however, none of them has successfully reached the combined

best efficiency-to-cost ratio. For instance, an English company has developed a smart walking cane called "WeWalk Smart Cane", as shown in Fig(1), that aims to help visually impaired people navigate. The project has some advantages as it is capable of detecting obstacles using ultrasonic sensors.



However, it has drawbacks as it only supports Turkish and English languages and is a bit expensive. Moreover, another solution has been offered by OrCam. They have invented an extension called "OrCam MyEye" that can be attached to any normal glasses. It can read text, identify faces, and help visually impaired people live a more independent life in general. Although it might seem to be the ultimate solution to all blind people's issues, it has some serious disadvantages. First, this device is unbelievably expensive costing about 3500\$. Second, its battery only lasts for two hours.

Smart glasses were manufactured that are capable of reading texts, detecting objects, and measuring distances, also all these data transferred to a spoken language in the headphone as an output. The main target was to present a solution with the highest possible accuracy at a much lower responding time. A test plan was conducted to measure the accuracy and the response time. It was concluded that our solution is reliable by meeting the two design requirements, which are accuracy of more than 85% and time responding between 3 and 5. , as the glasses achieve a 4-second responding time, while the accuracy of reading text is 97%, object detection accuracy with 87.5%, and object recognition 98% with 4.5 second responding time. This was mainly achieved by depending only on local materials discussed in the next section.

Materials



Method

1. The code was constructed with Python programming language on the visual studio, fig (2), developed for many times, and tested to see if there were any errors.
2. The Raspberry OS was downloaded and burned to the memory card through any computer.
3. The memory card was put into the Raspberry Pi, and the Raspberry Pi was later connected to a power bank, a monitor, a keyboard, and a mouse, as shown in fig(3)
4. The Raspberry was turned on and set up to be ready to read Python language, then the code was downloaded.
5. The two ultrasonic sensors and the buttons were connected to the breadboard by jumper wires, then the breadboard was connected to the raspberry by a jumper wires, fig (4).
6. The components were later fixed to the VR goggles to be easier for the visually impaired person to walk freely with the prototype.
7. A mobile phone was connected to the Raspberry by via Bluetooth to be used as an optical sensor for object and text detection.
8. The headset was connected to the Raspberry through the audio jack so the visually impaired person can hear the alerts of near obstacles and the audio of any written text around him.
9. The prototype was later trained on models of common objects and written texts to be able to identify anything that might confront the visually impaired person.
10. Finally, the prototype was tested again to see if everything worked correctly, and the connections are valid

After we had finished our prototype, we started doing our test plan, to make sure of our idea and to ensure we would achieve our desire. So, some steps were made for the test:

- 1) scenario simulating various environments was created (indoor, outdoor, crowded spaces) to evaluate the smart glass's ability to detect and recognize objects.
- 2) Smart glass's text recognition capabilities were tested with different fonts, sizes, and styles to ensure versatility.

The accuracy of the prototype was measured by exposing it to several real-life situations where it confronts unknown text passages/objects:

- 1) Analyzing how many words/objects it read correctly.
- 2) The accuracy percentage was calculated using the following equation:

$$\text{Accuracy} = \frac{\text{The no. of Detected words or objects}}{\text{The total no.}} \times 100$$

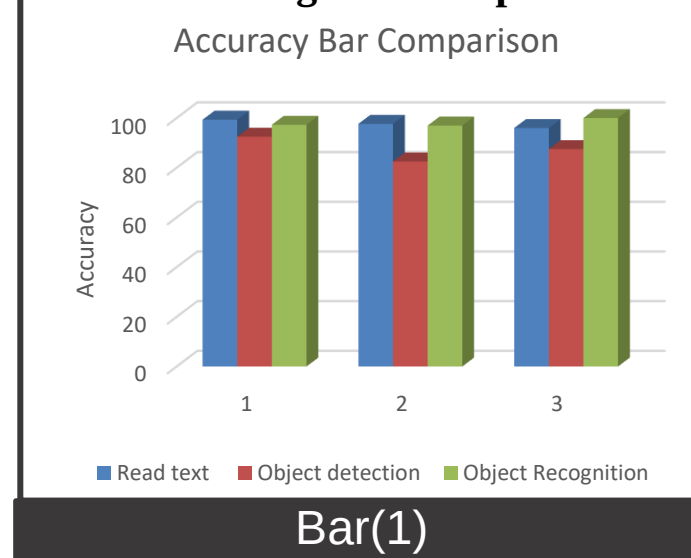
Also, the stopwatch was used to calculate and estimate the responding time for both OCR and object recognition.

Results

After testing the prototype for many times, results were recorded and both negative and positive results appeared

Negative results:

The first time constructing the prototype, the results were unsatisfying due to some construction issues as there were some incorrect wiring connections in the prototype. After those connections were modified as they have been written in the methods section, positive results began to appear as shown in the following graphs and calculations.



Positive results :

The prototype had achieved an average accuracy percentage of 97.66% in reading text, 87.5% in detecting objects and 98.1% in recognizing them. Also, the prototype achieved a lower cost of 81.5% compared to its global competitor (weWALK smart cane).

TRIAL	CORRECTLY DETECTED WORDS	TOTAL WORDS	ACCURACY PERCENTAGE	Time Response
1 ST TRIAL	288	300	96%	4.31s
2 ND TRIAL	293	300	97.6667%	3.92s
3 RD TRIAL	298	300	99.3334%	3.81s
Average	293	300	97.6%	4.01s

TRIAL	CORRECTLY DETECTED OBJECTS	TOTAL OBJECTS	DETECTION ACCURACY	CORRECTLY RECOGNIZED	RECOGNITION ACCURACY	Time response(s)
1 ST TRIAL	35	40	87.5%	35	100%	4.40 s
2 ND TRIAL	33	40	82.5%	32	96.97%	4.62s
3 RD TRIAL	37	40	92.5%	36	97.29%	4.48s
Average	35	40	87.5%	34	98%	4.5s

Analysis

The World Health Organization (WHO) estimated that globally, approximately 45 million people are blind. In comparison, around 285 million people have moderate to severe visual impairment. About 1.3 million people in Egypt had moderate to severe vision impairment. Therefore, the project will show a great demand due to the high portion affected by this ordeal. Leading to operating the factories to keep up with improvements and the latest version of this primary model. In addition, facilities served to patients ensure equality, deal with various situations, and accomplish tasks. Preserving public health and urging industrialization. Moreover, patients could be a practical part of the community that could help in industry development.

Computer vision is a field of artificial intelligence that trains computers to interpret and understand the visual world. Using digital images from cameras and videos and deep learning models, machines can accurately identify and classify objects and then react to what they "see." Along the same lines, our smart glasses work perfectly. The process's input is a digital image captured by the optical sensor, which is the camera of the user's smartphone. The mobile phone is connected directly to the Raspberry Pi via Bluetooth (PH.3.04&PH.3.05) *. The Raspberry Pi starts to process and analyze the image, state what object or text is in the image, and then provide auditory outputs to the user through a headset connected to the Raspberry Pi. We discuss artificial neural networks in the following paragraphs to better understand how all these components are connected. After numerous improvements, the accuracy of reading text is 97%, object detection accuracy with 87.5%, and object recognition 98% with 4.5 second responding time. More than hoped design requirement (which are accuracy of more than 85% and time responding between 3 and 5. , as the glasses achieve a 4-second responding time)

Artificial Neural Networks: What, How and Why?

First: What Neural Networks Are:

As the name suggests, neural networks simulate what science knows about human brain function. It is a computing unit formed by a combination of small but highly interconnected elements called "neurons". Neurons are units that hold a specific value and are arranged in distinct layers. The first layer contains the input neurons, the pixels of an image for example. The last layer is a combination of all the possible outcomes, possible predictions of What the image can be. The layers in between are called "hidden layers", "where all the processing happens. The number of hidden layers differs from one system to another according to the variety of inputs and their probabilities. Each layer is connected to the next layer through so-called "channels." Channels are characterized by weights and biases (commonly referred to as "W" and "b"). Weights are random real values given to each input neuron to express how vital the corresponding input is to the output. Meanwhile, biases are constants that act as intercepts in linear functions. They aim to give the model more accuracy and flexibility by providing a wider variety of variables to modify. The system adjusts weights and biases to reach the most perfect combination that suits the outputs. This structure, shown in fig(5) introduces inputs to the computing unit in the first layer. After that, it goes through multiple processes before coming up as outputs fig(5). But how are our inputs changed to be output?

Second: How Neural Networks Work:
For inputs to be outputs, they must undergo some processing. When an input reaches the hidden layers, it passes through what is called an activation function (sigmoid function). The sigmoid function $\frac{1}{1+e^{-x}}$ tests whether an input is active or not. It takes in $\sum (w \cdot x + b)$ and outputs a value between 1 and 0 (MA.3.01) *, where more negative values approach 0, thus considered inactive, whereas more positive values approach 1, thus considered active as shown in Fig(6). Generally, the output of the activation function in one hidden layer is given as an input to the next neuron in the next hidden layer. This process called forward propagation, continues to happen until the system reaches the output layer.

Third: Why Neural Networks are able to learn:
The gripping strength of machine learning algorithms is their capability to develop and give improved outputs every time. All machine learning models experience a training phase in which the system compares the real or desired output of the training inputs with the predicted output. During this phase, the model adjusts the values of weights and biases go through a process called backward propagation. The error is calculated in this process, and the propagation starts from the output layer as shown in Fig(7). While moving from one layer to another, weights and biases change until the input layer is reached. Then, the model tests the training inputs again. The process continues until the output from the network approximates the real value for all the training inputs.

Ultrasonic Sensors:
In addition to detecting texts and images, the smart glasses can alarm the user if any obstacles are on his road, and this would have never been achieved without attaching an ultrasonic Sensor. It is a device by which we can measure the distance to an object using ultrasonic sound waves (PH.3.01). Generally, an ultrasonic sensor uses transducers to transmit an ultrasonic wave and receive its echo. After that, the sensor can determine the distance to the target by measuring the time lapse between the sent and the received ultrasonic pulse. In summary, an ultrasonic sensor follows the rule in Fig (8) when measuring the distance to a specific target.

Programming and Libraries:
First, OCR:
Python-tesseract is an optical character recognition (OCR) tool for Python. It will recognize and "read" the text embedded in images. Python-tesseract is a wrapper for Google's Tesseract-OCR Engine. It is also helpful as a stand-alone invocation script to Tesseract, as it can read all image types supported by the Pillow and Leptonica imaging libraries. Meanwhile, OCR accuracy depends on the text's size and distance.

Second, Object Detection:
The COCO dataset used: The COCO (Common Objects in Context) dataset classes are divided into two main categories: "things" and "stuff." Also, the accuracy of detection depends on video resolution "Things" classes include objects easily picked up or handled, such as animals, vehicles, and household items. Examples of "things" classes in COCO are Person, Bicycle, Car, and Motorcycle. "Stuff" classes include background or environmental items such as sky, water, and road. It contains over 330,000 images, each annotated with 80 object categories.

Learning outcomes:
PH.3.01 This LO discusses waves, their types and how they travel in different mediums. This LO discusses the fundamentals of communication and important vocabulary It helped us understand what type of waves ultrasonic sensors depend on. It helped us understand how we can establish a connection between our smart glasses and the user PH.3.05 This LO is a sequel to the previous one. It discusses the concept in further details adding some real-life applications. This LO discusses functions, their derivatives, and properties. It helped us understand how the connection is established through Wi-Fi and Bluetooth. After studying this LO, we could know more about what sigmoid functions are and how they work.

Conclusion

According to scientific analysis, extended research, dedicated work, and reasoning, the prototype is a reasonable and effective solution for the main problem which is the disability of blindness. The prototype is capable of solving the problem because of its high accuracy, efficiency, suitable cost, highly acceptable responding time, and eco-friendly The project was deduced to be usable as a smart device to help visually impaired people. In conclusion, the prototype achieved the design requirements of high accuracy in detecting reading texts, and identifying objects and possible obstacles, and this is proven by the extraordinary results produced by the Prototype in small responding time.

Recommendation

GPS Navigation:

GPS work using triangulation interaction between three or more satellites. Each GPS satellite sends a great sign and orbital boundaries that permit GPS gadgets to interpret and process the satellite's precise area. According to our smart glasses, GPS technology furnishes the disabled, visually impaired one with the easiest method of getting to any location he needs. This Technology will be actualized by providing the glasses with an informational set of data from Google Maps about all shops, clubs, or restaurants close to where the patient lives. Besides, the patient can likewise effectively set up his current co-ordinates and offer to his contacts.

Voice Assistance:

The speech recognition system receives the voice, and turns it into Text by separating a discourse recording into singular sounds, examining each sound, and utilizing logarithms to locate the most likely The word fits in that language and deciphering those sounds into Text. This Text can be used as an input to the smart glasses. The Text is put under processing called natural language processing (NLP); via AI, the intelligent device does what the user orders accurately, correctly, and faster than humans.

Phonak Roger Pen:

The Roger Pen is a portable microphone used in conjunction with the hearing aids, It allows the user to hear and understand sounds in noisy places and distant speech. It instantly transmits the signals via high-quality sound and can connect to cell phones and other audio outputs for TV and multimedia listening through Bluetooth. This device could be used along with a speech recognition system for the ultimate voice assistance experience.

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